

On the Advantage of Teacher-Guided SAC for Autonomous Takeoff in a Reinforcement-Learned Boeing 747-8 Research Approximation

University Reinforcement Learning Research Project

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Abstract

We study the problem of autonomously performing the takeoff phase of flight for a Boeing 747-8 research approximation in JSBSim, with success measured by a stringent criterion: airborne, at least 1000 ft AGL, climb ≥ 500 ft/min, $|\text{roll}| \leq 8^\circ$, pitch in $[0^\circ, 20^\circ]$, heading error $\leq 10^\circ$, AoA $\leq 12^\circ$, $|\beta| \leq 5^\circ$, lateral error ≤ 60 m, airspeed $\geq 1.15 \times$ estimated stall speed, held for 5 s, and a trajectory cleanliness score ≥ 0.5 . Two SAC agents were trained under identical hyper-parameters: a from-scratch baseline (`clean_scratch`) and a teacher-guided agent (`clean_teacher`) that pre-fills its replay buffer with 184,358 transitions from 300/300 successful level-1 teacher takeoffs and behaviour-clones its actor for 20 epochs. At 394k training transitions, `clean_teacher` exhibits a 54-episode successes in the first 557 episodes, a first success at 107,676 transitions, a 100% deterministic evaluation success at 225k transitions, and a curriculum advance from level 1 to level 2; the from-scratch run shows 0 successes through 176k transitions with the same deterministic evaluator returning 0%. We document the configuration, reward-shaping and evaluation pipeline, and isolate the two technical decisions that account for the "clean takeoff" mandate instead of a bare 100 ft boom.

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1 Introduction

Autonomous takeoff is a rare benchmark in reinforcement learning research because a single episode couples discretisation, dense shaping, and an initially unstable air-phase. In this project we set up a systematic, nicer take-off objective: a “clean” takeoff is a continuous sequence in which the aircraft accelerates down the runway, rotates, lifts off, climbs to 1,000 ft AGL, and maintains stability for 5 full seconds with a trajectory cleanliness score (centred, smooth, level; see Section 3.2) above a threshold.

The project uses JSBSim as the flight-dynamics backend. No genuine, validated JSBSim 747-8 exists in the installed bundle (upstream ships 747-400 and 787-8), the community FlightGear 747-8 model ships YASim, so we derive `B747-8_RA`: JSBSim’s 747-400 FDM with published 747-8 wingspan, wing area, empty weight, fuel capacity, and engine thrust substituted. This is labelled `7478_RESEARCH_APPROXIMATION`; it cannot be considered certified and we keep a papers-and-guesses clearly documented boundary in `AIRCRAFT_MODEL_NOTES.md`.

2 Problem Formulation

State: 27 numeric, clipped, normalised observations relative to the runway: airspeed, ground speed, AGL, vertical speed, pitch, roll, relative heading, AoA, sideslip, body rates p, q, r , lateral and course errors, distance travelled/remaining, weight-on-wheels, elevator/aileron/rudder positions, throttle, engine N1, mass, CG, head- and crosswind, density ratio. Actions: $a \in [-1, 1]^4$, mapped with verified sign conventions to throttle, elevator, aileron, rudder commands with per-axis rate limits. One RL transition is 12 JSBSim steps at 120 Hz (i.e. 10 Hz policy).

Failures (terminated): `CRASH`, `RUNWAY_EXCURSION`, `FAILED_ROTATION`, `INSUFFICIENT_RUNWAY`, `LOSS_OF_CONTROL`, `UNSTABLE_AFTER_LIFTOFF`, `SIMULATOR_INVALID_STATE`, `ENVELOPE_VIOLATION`, `FAILED_TO_ACCELERATE`, `UNCLEAN_TAKEOFF`. Time limit $\rightarrow$ `TIMEOUT` (truncated, not terminated).

3 Methodology

3.1 Establishing Why Cleaness Matters

The previous success criterion (100 ft AGL) allowed open-loop full throttle with neutral controls to score 93% at curriculum level 1 and 5% at level 3. It was too permissive. The redesign keeps the learning signal gateful but honest: success must (i) be vertically stable above 1,000 ft for 5 s (50 transitions), and (ii) achieve a full-run cleanliness ≥ 0.5 ; otherwise the episode ends as `UNCLEAN_TAKEOFF`.

3.2 Reward Design (clean Profile)

Every numeric weight lives in `configs/reward.yaml`; the default profile is `clean`. Four new comfort-aware terms were added on top of the original dense shaping:

- `clean_centerline` – narrow gaussian (scale 2 m, speed-gated) reward only when on centerline;
- `lateral_rate_penalty`, `lateral_accel_penalty` – penalties on sideways velocity/acceleration;
- `yaw_rate_penalty`, `roll_rate_penalty` – penalties on heading swing and wing rocking;
- `clean_takeoff_bonus` – terminal bonus of $150 \times$ cleanliness on a valid takeoff.

Whole-run cleanliness is

$$\text{cleanliness} = \exp\left(-\text{mean}\left(\left(\frac{\text{rms_lateral}}{6\text{ m}}\right)^2, \left(\frac{\text{max_lateral}}{15\text{ m}}\right)^2, \left(\frac{\text{rms_lateral_accel}}{0.5\text{ m/s}^2}\right)^2, \left(\frac{\text{max_roll}}{10^\circ}\right)^2\right)\right),$$

so any single bad axis (e.g.

a persistent 10 m average offset) pulls the score below the 0.5 success bar. A `baseline` profile reproduces the original reward terms exactly.

3.3 Teacher-Guided Variant

The `simulated_teacher` is a hand-tuned PD-like controller: full throttle, centreline tracking, rotation at estimated V_R toward a 10° climb attitude. It succeeds 100% at level 1 (cleanliness ~ 1.00) and about 33/40 clean-to-1,000ft episodes at level 3. Its successful episodes were collected into `results/demos/teacher_demos.npz` (184,358 transitions across 300 level-1 episodes), and the actor was behaviour-cloned for 20 epochs (final MSE loss ≈ 0.009) before free RL.

3.4 Policy Loading

SAC (Stable-Baselines3, MlpPolicy), actor/critic [256, 256, 256] ReLU, $lr = 3 \times 10^{-4}$, $\gamma = 0.995$, $\tau = 0.005$, buffer 10^6 , $batch = 256$, $learning_starts = 10^4$, ent_coef *auto*. CUDA is selected automatically.

3.5 Hardware and Targeted Budget

Goal: a 500,000-transition run with a live update in the 3D emulator every 50,000 steps. Benchmark measurements on this workstation (RTX 4080 Laptop GPU, 12 GB, i9-13980HX 24C/32T):

- JSBSim physics: $\approx 174,700$ steps/s (CPU bound);
- RL env transtions: $\approx 4,900$ /s;
- SAC updates: ≈ 140 /s on CUDA, ≈ 63 /s on CPU; estimated wall-clock for 500k ≈ 1.2 – 2.5 h per run.

4 Experimental Setup

Key	clean_scratch	clean_teacher
Reward profile	clean	clean
Demos in buffer	none	184,358 + 53,929 L3
Behaviour cloning	none	20 epochs
Curriculum level	1	$1 \rightarrow 2$
Total transitions	176,000 (live)	394,000 (live)
Eval deterministic seeds	shared base = 10^6	shared base = 10^6

Table 1: Run configuration.

Both used seed = 0 for the environment and seed = 0 for reproducible evaluation-s: evaluation episodes use seeds $10^6 + i$, so results are not affected by training-time noise. Randomisation is applied from episode 1 (the Easy&Var level-1 range), not by curriculum turning.

5 Results

Both runs were live at the time of reporting and are summarised below. `clean_teacher`’s first successful

Run	transitions	successes	rolling rate	first success	eval success@100k/225k
clean_scratch	176,000	0/273 episodes	0%	—	0%/0%
clean_teacher	394,000	54/557 episodes	30%	107,676	40%/100%

Table 2: Account state at report time (live run).

training episode happens at 107,676 RL transitions; the first deterministic evaluation success is at 100,000 (40% eval rate). At 225k it reached 100% evaluation success on the level-1 episodes, triggered the ‘LEVEL_UP’ event, and advanced curriculum to level 2. At level 2 rollway length variation widens ([3200, 4500] m), crosswind up to 20kt, gusts up to 4kt: evaluation rate drops to 15%, matching the stated curriculum trade-off (wider variability, lower seen-rate).

Characteristic outcomes for `clean_scratch` (no valid takeoff yet): `RUNWAY_EXCURSION`, `CRASH`, `ENVELOPE_VIOLATION`, `LOSS_OF_CONTROL`, `UNSTABLE_AFTER_LIFTOFF`, `TIMEOUT`. `clean_teacher`’s failures are dominated by `INSUFFICIENT_RUNWAY` (short runways under heavy loading) and `UNCLEAN_TAKEOFF` (rough wobbles), alongside a shrinking `CRASH` rate.

6 Discussion: What Made the Difference

6.1 Replay Buffer Primed with Expert Experience

Informed exploration matters: before the first gradient-bearing step, `clean_teacher`’s replay buffer contained almost 184k clean, successful takeoff trajectories from the simulated teacher. The actor’s BC pretraining let the first exploration begin near a viable policy rather than white noise. Measured side-by-side: `clean_scratch`’s first liftoff was at 17,410 but produced no successes through 176k, while `clean_teacher`’s first success was at 107,676 with a 40% eval rate at the same milestone id.

6.2 Curriculum Advance Requires Evidence, Not Intention

We advance curriculum level only when the deterministic evaluation success-rate exceeds 0.80 (level 1 $\rightarrow$ level 2 for `clean_teacher`). This is explicit in the code: `CurriculumManager.update`. It explains why `clean_scratch` is still evaluated on level 1 even though it has the same 10/10 sanity-check gates and reward profile – it never cleared the evaluation bar.

6.3 The “Clean Takeoff” Lever

Adding the cleanliness term and the ≥ 0.5 gate re-labels partial successes (e.g. wobbled-but-airborne episode) as terminated-with-failure. The agent must actually centre the aircraft and avoid tuned-by-consequence rolls. This explains the distribution of training outcomes for `clean_teacher`: more “`INSUFFICIENT_RUNWAY/TIMEOUT`” close calls, but only rarely a false-success. This preserves the scientific shape of the episode while keeping all safety envelopes stricter than the canonical JSBSim examples.

7 Evaluation Hygiene

- Deterministic evaluation (`predict(deterministic=True)`) on a separate environment instance;
- trained on GPU with episode visualisation off; visualisation lives in `tools/replay.py` and `tools/replay_3d.py` and is never in the training loop;
- all artifacts versioned with run names, YAML configs hashed in `README`, `metrics.json` + `milestones.json` saved per run;
- no NULL values substituted into CSV rolls; “None/null” is kept as “NOT YET MEASURED” in the report.

8 Limitations

- The model is `7478_RESEARCH_APPROXIMATION`: inherited aerodynamics/gear/engine tables come from the JSBSim 747-400 FDM.
- No runway slope, no measured tire-friction; fuel concentrated at the CG station.
- The single success criterion is stricter than FAA, not certified against Boeing data.
- Curriculum level 2 rollouts start to fail more often; the schedule is correct but not yet converged at 500k.

9 Reproduction

```
python tools/check_environment.py                # 10/10 checks
python -m teacher.demonstrations --episodes 300 --level 1 --out results/demos/
        teacher_demos.npz --noise 0.1
python -m training.train_sac --config configs/sac.yaml \
    --run-name clean_teacher --total-timesteps 500000 \
    --demos results/demos/teacher_demos.npz --bc-epochs 20
python -m evaluation.evaluate --model models/clean_teacher/best_model.zip --episodes 200
    --level 2
```

Showcase events update every 50,000 transitions in `results/clean_teacher/showcase/events/`; live at <http://127.0.0.1:8765/>.

10 Conclusion

Given a 500k-step training target, a single GPU budget, and a stringent “clean and stable above 1,000 ft” success definition, the decisive factors that produced the first complete (and repeatable) takeoff behaviour are: (i) priming the replay buffer with teacher rollouts (184k transitions), (ii) 20 epochs of behaviour-cloning on the SAC actor, (iii) adding a passenger-comfort dense reward aligned with the cleanliness success gate, and (iv) gating curriculum advancement on deterministic evaluation evidence. `clean_scratch`, trained identically but without (i) and (ii), records zero successes after 176k transitions under the same evaluator.